Type MS · Angle between two sloping faces with the cosine rule

Answer 1 of 9 · Question 8 part e · 3 marks

2025 JUN · Paper 01 · extract 1 of 4 · find part e in this mark scheme extract

8(a)	$\tan 30° = \frac{VP}{6}$ where P is the centre of the base $VP = 6\tan 30° = 2\sqrt{3}$ *	M1 A1*cso [2]
(b)	$AC = \sqrt{12^2 + 12^2} = 12\sqrt{2}$ or $AC = \frac{12}{\sin 45}$ or $AC = \frac{12}{\cos 45}$ $VA^2 = \left(2\sqrt{3}\right)^2 + \left(6\sqrt{2}\right)^2 (= 84)$ $VA = 2\sqrt{21}$ (cm) oee	M1 oe M1 oe A1 [3]
(c)	$VX = \sqrt{\left('2\sqrt{21}'\right)^2 - 6^2} \left(= 4\sqrt{3}\right)$ or $VX = \frac{6}{\cos 30}$ or $VX = \sqrt{6^2 + \left(2\sqrt{3}\right)^2}$ where X is the midpoint of the side AD Area $= \frac{1}{2} \times 4\sqrt{3} \times 12 = 24\sqrt{3}$ (cm^2) oee	M1 M1A1 [3]
(d)	$'24\sqrt{3}' = \frac{1}{2} \times '2\sqrt{21}' \times DY$ where Y is the foot of the perpendicular from D to VA $DY = \frac{24\sqrt{7}}{7}$ oee	M1 A1 [2]
(e)	$\cos\theta = \frac{\left(,\frac{24\sqrt{7}}{7},\right)^2 + \left(,\frac{24\sqrt{7}}{7},\right)^2 - \left('12\sqrt{2}'\right)^2}{2 \times ,\frac{24\sqrt{7}}{7}, \times ,\frac{24\sqrt{7}}{7},} = -\frac{3}{4}$ $\theta = 138.6°$	M1A1 A1 [3]
		Total is 13 marks

Type MS · Angle between two sloping faces with the cosine rule

Answer 1 of 9 · Question 8 part e · 3 marks

2025 JUN · Paper 01 · extract 2 of 4 · find part e in this mark scheme extract

Part	Mark	Additional Guidance
(a)	M1	For forming a correct equation using trigonometry. e.g. $\tan 60° = \frac{6}{VP}$ or $\frac{h}{\sin 30°} = \frac{6}{\sin 60°}$ (If attempted by ratio, send to review)
	A1*cso	For **rearranging** to show the required result. Minimum need to see one intermediate step from their equation to the answer ,if the height is not the subject of their equation: e.g. $\tan 30° = \frac{VP}{6} \Rightarrow VP = 6\tan 30° = 2\sqrt{3}$ is M1A1 $VP = 6\tan 30° = 2\sqrt{3}$ is M1A1 $\tan 30° = \frac{VP}{6} \Rightarrow \frac{\sqrt{3}}{3} = \frac{VP}{6} \Rightarrow VP = 2\sqrt{3}$ is M1A1 $\tan 30° = \frac{VP}{6} \Rightarrow VP = 2\sqrt{3}$ is M1A0
(b)	M1oe	Correct use of Pythagoras' theorem or trigonometry to obtain the length/half length of the diagonal of the base AC or $\frac{1}{2}AC$ Or other valid method to find VM, (Let M to be the midpoint of AB) e.g.: $VM = \frac{6}{\cos 30}\left(= 4\sqrt{3}\right)$ or $VM = \sqrt{6^2 + \left(2\sqrt{3}\right)^2}\ \left(= 4\sqrt{3}\right)$
	M1oe	Use Pythagoras' theorem to obtain a numerical value for VA / VA^2 Any correct method. e.g. $VA^2 = \left(4\sqrt{3}\right)^2 + (6)^2$
	A1	Correct exact length $2\sqrt{21}$ or $\sqrt{84}$ oee (or exact equivalent)
(c)	M1	Correct use of Pythagoras' theorem or trigonometry to obtain the length of VX, which is the same as VM, could be found in part b Or correct use of cosine rule (ft their VA) to find the value of cosine value of any angle in triangle VAD e.g. $AD^2 = VA^2 + VA^2 - 2VA \times VA \times \cos \angle AVD$ $12^2 = \left('2\sqrt{21}'\right)^2 + \left('2\sqrt{21}'\right)^2 - 2 \times '2\sqrt{21}' \times '2\sqrt{21}' \times \cos \angle AVD$ $\left\{\Rightarrow \cos \angle AVD = \frac{1}{7}\right\}$
	M1	Correct method to find the area of triangle VAD (Do not have to use exact values) Accept any correct valid method. e.g. $\left\{\cos \angle AVD = \frac{1}{7} \Rightarrow \sin \angle AVD = \frac{4\sqrt{3}}{7}\right\} \Rightarrow A = \frac{1}{2} \times '2\sqrt{21}' \times '2\sqrt{21}' \times '\frac{4\sqrt{3}}{7}' = 24\sqrt{3}$
	A1	Correct **exact** value for area
(d)	M1	Use of area of triangle with VA as the base to form an equation for DY Or other valid method: e.g. using trigonometry ratio: $\sin \angle VAD = \frac{'4\sqrt{3}'}{'2\sqrt{21}'} = \frac{DY}{12}$

Type MS · Angle between two sloping faces with the cosine rule

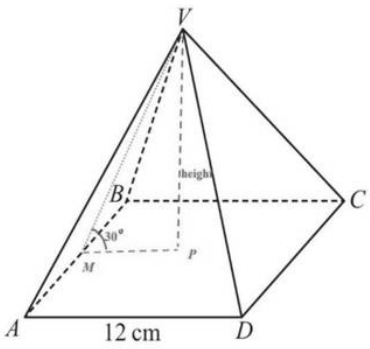
Answer 1 of 9 · Question 8 part e · 3 marks

2025 JUN · Paper 01 · extract 3 of 4 · find part e in this mark scheme extract

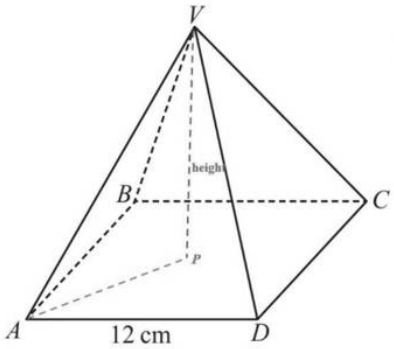
	A1	Correct exact length $DY = \frac{24\sqrt{7}}{7}$ oee, e.g. $DY = \frac{8\sqrt{63}}{7}$ or $24\sqrt{\frac{1}{7}}$
(e)	M1	Correct use of cosine rule: $\cos\theta = \frac{"DY"^2 + BY"^2 - "BD"^2}{2\times"DY"\times"BY"}$ with their values Or other valid method, e.g. $\sin\frac{\theta}{2} = \frac{"\frac{BD}{2}"}{"DY"} = \frac{"6\sqrt{2}"}{"\frac{24\sqrt{7}}{7}"}$
	A1	Obtains $\cos\theta = -\frac{3}{4}$ or $\sin\frac{\theta}{2} = \frac{\sqrt{14}}{4}$
	A1	Correct angle found

Useful diagrams:

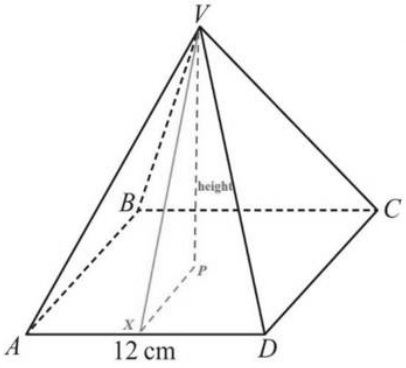
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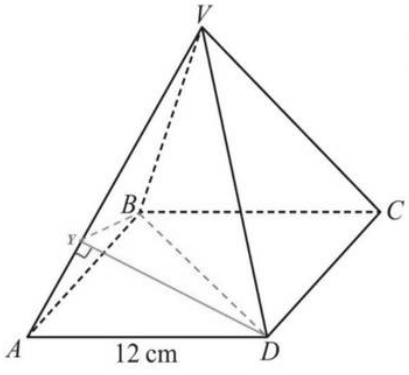
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4



Type MS · Angle between two sloping faces with the cosine rule

Answer 1 of 9 · Question 8 part e · 3 marks

2025 JUN · Paper 01 · extract 4 of 4 · find part e in this mark scheme extract

Question number	Scheme	Marks

Type MS · Angle between two sloping faces with the cosine rule

Answer 2 of 9 · Question 8 part d · 2 marks

2025 JUN · Paper 01 · extract 1 of 4 · find part d in this mark scheme extract

8(a)	$\tan 30° = \frac{VP}{6}$ where P is the centre of the base $VP = 6\tan 30° = 2\sqrt{3}$ *	M1 A1*cso [2]
(b)	$AC = \sqrt{12^2 + 12^2} = 12\sqrt{2}$ or $AC = \frac{12}{\sin 45}$ or $AC = \frac{12}{\cos 45}$ $VA^2 = \left(2\sqrt{3}\right)^2 + \left(6\sqrt{2}\right)^2 (= 84)$ $VA = 2\sqrt{21}$ (cm) oee	M1 oe M1 oe A1 [3]
(c)	$VX = \sqrt{\left('2\sqrt{21}'\right)^2 - 6^2} \left(= 4\sqrt{3}\right)$ or $VX = \frac{6}{\cos 30}$ or $VX = \sqrt{6^2 + \left(2\sqrt{3}\right)^2}$ where X is the midpoint of the side AD Area $= \frac{1}{2} \times 4\sqrt{3} \times 12 = 24\sqrt{3}$ (cm^2) oee	M1 M1A1 [3]
(d)	$'24\sqrt{3}' = \frac{1}{2} \times '2\sqrt{21}' \times DY$ where Y is the foot of the perpendicular from D to VA $DY = \frac{24\sqrt{7}}{7}$ oee	M1 A1 [2]
(e)	$\cos\theta = \frac{\left(,\frac{24\sqrt{7}}{7},\right)^2 + \left(,\frac{24\sqrt{7}}{7},\right)^2 - \left('12\sqrt{2}'\right)^2}{2 \times ,\frac{24\sqrt{7}}{7}, \times ,\frac{24\sqrt{7}}{7},} = -\frac{3}{4}$ $\theta = 138.6°$	M1A1 A1 [3]
		Total is 13 marks

Type MS · Angle between two sloping faces with the cosine rule

Answer 2 of 9 · Question 8 part d · 2 marks

2025 JUN · Paper 01 · extract 2 of 4 · find part d in this mark scheme extract

Part	Mark	Additional Guidance
(a)	M1	For forming a correct equation using trigonometry. e.g. $\tan 60° = \frac{6}{VP}$ or $\frac{h}{\sin 30°} = \frac{6}{\sin 60°}$ (If attempted by ratio, send to review)
	A1*cso	For **rearranging** to show the required result. Minimum need to see one intermediate step from their equation to the answer ,if the height is not the subject of their equation: e.g. $\tan 30° = \frac{VP}{6} \Rightarrow VP = 6\tan 30° = 2\sqrt{3}$ is M1A1 $VP = 6\tan 30° = 2\sqrt{3}$ is M1A1 $\tan 30° = \frac{VP}{6} \Rightarrow \frac{\sqrt{3}}{3} = \frac{VP}{6} \Rightarrow VP = 2\sqrt{3}$ is M1A1 $\tan 30° = \frac{VP}{6} \Rightarrow VP = 2\sqrt{3}$ is M1A0
(b)	M1oe	Correct use of Pythagoras' theorem or trigonometry to obtain the length/half length of the diagonal of the base AC or $\frac{1}{2}AC$ Or other valid method to find VM, (Let M to be the midpoint of AB) e.g.: $VM = \frac{6}{\cos 30}\left(=4\sqrt{3}\right)$ or $VM = \sqrt{6^2 + \left(2\sqrt{3}\right)^2}\ \left(=4\sqrt{3}\right)$
	M1oe	Use Pythagoras' theorem to obtain a numerical value for VA / VA^2 Any correct method. e.g. $VA^2 = \left(4\sqrt{3}\right)^2 + (6)^2$
	A1	Correct exact length $2\sqrt{21}$ or $\sqrt{84}$ oee (or exact equivalent)
(c)	M1	Correct use of Pythagoras' theorem or trigonometry to obtain the length of VX, which is the same as VM, could be found in part b Or correct use of cosine rule (ft their VA) to find the value of cosine value of any angle in triangle VAD e.g. $AD^2 = VA^2 + VA^2 - 2VA \times VA \times \cos\angle AVD$ $12^2 = \left('2\sqrt{21}'\right)^2 + \left('2\sqrt{21}'\right)^2 - 2\times'2\sqrt{21}'\times'2\sqrt{21}'\times\cos\angle AVD$ $\left\{\Rightarrow \cos\angle AVD = \frac{1}{7}\right\}$
	M1	Correct method to find the area of triangle VAD (Do not have to use exact values) Accept any correct valid method. e.g. $\left\{\cos\angle AVD = \frac{1}{7} \Rightarrow \sin\angle AVD = \frac{4\sqrt{3}}{7}\right\} \Rightarrow A = \frac{1}{2}\times'2\sqrt{21}'\times'2\sqrt{21}'\times'\frac{4\sqrt{3}}{7}' = 24\sqrt{3}$
	A1	Correct **exact** value for area
(d)	M1	Use of area of triangle with VA as the base to form an equation for DY Or other valid method: e.g. using trigonometry ratio: $\sin\angle VAD = \frac{'4\sqrt{3}'}{'2\sqrt{21}'} = \frac{DY}{12}$

Type MS · Angle between two sloping faces with the cosine rule

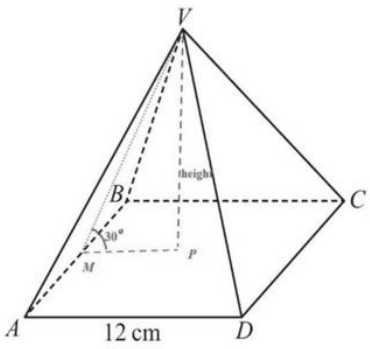
Answer 2 of 9 · Question 8 part d · 2 marks

2025 JUN · Paper 01 · extract 3 of 4 · find part d in this mark scheme extract

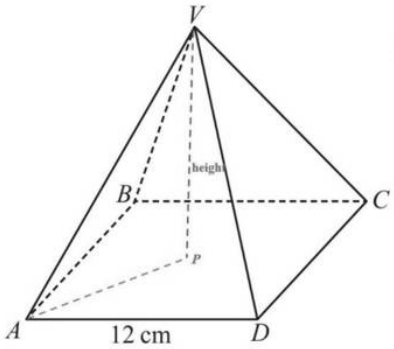
	A1	Correct exact length $DY = \frac{24\sqrt{7}}{7}$ oee, e.g. $DY = \frac{8\sqrt{63}}{7}$ or $24\sqrt{\frac{1}{7}}$
(e)	M1	Correct use of cosine rule: $\cos\theta = \frac{"DY"^2 + BY"^2 - "BD"^2}{2\times"DY"\times"BY"}$ with their values Or other valid method, e.g. $\sin\frac{\theta}{2} = \frac{"\frac{BD}{2}"}{"DY"} = \frac{"6\sqrt{2}"}{"\frac{24\sqrt{7}}{7}"}$
	A1	Obtains $\cos\theta = -\frac{3}{4}$ or $\sin\frac{\theta}{2} = \frac{\sqrt{14}}{4}$
	A1	Correct angle found

Useful diagrams:

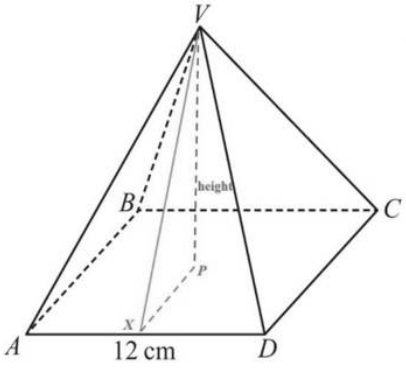
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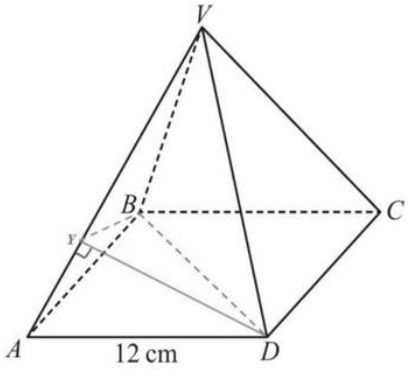
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3



4



Type MS · Angle between two sloping faces with the cosine rule

Answer 2 of 9 · Question 8 part d · 2 marks

2025 JUN · Paper 01 · extract 4 of 4 · find part d in this mark scheme extract

Question number	Scheme	Marks

Type MS · Angle between two sloping faces with the cosine rule

Answer 3 of 9 · Question 11 part c · 3 marks

2019 JUN · Paper 02 · extract 1 of 3 · find part c in this mark scheme extract

11(a)	$AC=\sqrt{\left(16^2+16^2\right)}=16\sqrt{2}$	M1A1
ALT:	$AP^2+PD^2=16^2\Rightarrow AP=8\sqrt{2}$	
	$VP=8\sqrt{2}\tan 45=8\sqrt{2}$ (where P is the centre of the base) *	M1A1cso (4)
(b)	$VA^2=\left(8\sqrt{2}\right)^2+\left(8\sqrt{2}\right)^2\left(=256\right)$ or $VA=\dfrac{8\sqrt{2}}{\sin 45°}$	M1A1
	$VA=16$ cm	A1 (3)
(c)	$DX^2=16^2-8^2$ where X is the foot of the perpendicular from D to VA	M1A1
	$DX=8\sqrt{3}$	A1 (3)
(d)	$\tan\theta=\dfrac{8\sqrt{2}}{8},\sin\theta=\dfrac{8\sqrt{2}}{8\sqrt{3}},\cos\theta=\dfrac{8}{8\sqrt{3}}$	M1A1
	(or unsimplified if cosine or sine rule used) $\theta=54.7°$	A1 (3)
(e)	$\cos\phi=\dfrac{\left(8\sqrt{3}\right)^2+\left(8\sqrt{3}\right)^2-\left(16\sqrt{2}\right)^2}{2\times8\sqrt{3}\times8\sqrt{3}}\left(=-\dfrac{1}{3}\right)$	M1A1
	$\phi=109.5°$	A1 (3) [16]

(a)	
M1	Use Pythagoras (with a + sign) to obtain the length of the diagonal of the base. Or use Pythagoras with correct sign to obtain the half diagonal
A1	Correct length for the diagonal or half diagonal
M1	Use tan in ΔAPV, their AP and angle of 45° to obtain the height
A1cso	Correct answer with no errors in the working
	OR: State ΔAVP is isosceles, or shown the 2 correct angles – can be on a diagram M1
	$VP\left(=AP\right)=8\sqrt{2}$ A1
	OR use any other complete valid method M1 Correct result A1
(b)	
M1	Use Pythagoras or trigonometry in ΔAPV or ΔAVC (or any other complete, valid method) to obtain a numerical expression for VA.
A1	Correct numbers in their choice of method.
A1	Correct length obtained.
	$AV=16$ w/o working scores M1A1A1

Type MS · Angle between two sloping faces with the cosine rule

Answer 3 of 9 · Question 11 part c · 3 marks

2019 JUN · Paper 02 · extract 2 of 3 · find part c in this mark scheme extract

Question Number	Scheme	Marks
(c)		
M1	Use Pythagoras with a minus sign (seen or implied) or trigonometry in ΔADX OR any other complete valid method **NB**: triangle ΔADV is equilateral.	
A1	Correct numbers in their choice of method.	
A1	Correct **exact** length for the perpendicular.	
(d)		
M1	Identify the correct triangle needed with the required angle marked (may be on Figure 1). This may be shown explicitly or implied by their work that follows.	
A1	Reach one of $\tan\theta = \frac{8\sqrt{2}}{8}$, $\sin\theta = \frac{8\sqrt{2}}{8\sqrt{3}}$, $\cos\theta = \frac{8}{8\sqrt{3}}$ oe	
A1	Correct answer, **must be 1 dp.**	
(e)		
M1	Use cosine rule: $\cos\phi = \frac{"DX"^2 + "XB"^2 - "BD"^2}{2\times"DX"\times"XB"}$ (their values)	
A1	Correct numbers substituted, follow through their previous answers	
A1	Correct answer, **must be 1 dp unless already penalised in (d)**	
	Any other routes should be marked: M1 Correct, complete method (ie it must be possible to reach a value for the required angle) A1 Correct numbers substituted A1 Correct answer, **must be 1 dp unless already penalised in (d)**	

Type MS · Angle between two sloping faces with the cosine rule

Answer 3 of 9 · Question 11 part c · 3 marks

2019 JUN · Paper 02 · extract 3 of 3 · find part c in this mark scheme extract

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Type MS · Angle between two sloping faces with the cosine rule

Answer 4 of 9 · Question 11 part e · 3 marks

2019 JUN · Paper 02 · extract 1 of 3 · find part e in this mark scheme extract

11(a)	$AC = \sqrt{(16^2 + 16^2)} = 16\sqrt{2}$	M1A1
ALT:	$AP^2 + PD^2 = 16^2 \Rightarrow AP = 8\sqrt{2}$	
	$VP = 8\sqrt{2}\tan 45 = 8\sqrt{2}$ (where P is the centre of the base) *	M1A1cso (4)
(b)	$VA^2 = \left(8\sqrt{2}\right)^2 + \left(8\sqrt{2}\right)^2 (= 256)$ or $VA = \dfrac{8\sqrt{2}}{\sin 45°}$	M1A1
	$VA = 16$ cm	A1 (3)
(c)	$DX^2 = 16^2 - 8^2$ where X is the foot of the perpendicular from D to VA	M1A1
	$DX = 8\sqrt{3}$	A1 (3)
(d)	$\tan\theta = \dfrac{8\sqrt{2}}{8}, \sin\theta = \dfrac{8\sqrt{2}}{8\sqrt{3}}, \cos\theta = \dfrac{8}{8\sqrt{3}}$	M1A1
	(or unsimplified if cosine or sine rule used) $\theta = 54.7°$	A1 (3)
(e)	$\cos\phi = \dfrac{\left(8\sqrt{3}\right)^2 + \left(8\sqrt{3}\right)^2 - \left(16\sqrt{2}\right)^2}{2\times 8\sqrt{3}\times 8\sqrt{3}} \left(= -\dfrac{1}{3}\right)$	M1A1
	$\phi = 109.5°$	A1 (3) [16]

(a)	
M1	Use Pythagoras (with a + sign) to obtain the length of the diagonal of the base. Or use Pythagoras with correct sign to obtain the half diagonal
A1	Correct length for the diagonal or half diagonal
M1	Use tan in ΔAPV, their AP and angle of 45° to obtain the height
A1cso	Correct answer with no errors in the working
	OR: State ΔAVP is isosceles, or shown the 2 correct angles – can be on a diagram M1
	$VP(= AP) = 8\sqrt{2}$ A1
	OR use any other complete valid method M1 Correct result A1
(b)	
M1	Use Pythagoras or trigonometry in ΔAPV or ΔAVC (or any other complete, valid method) to obtain a numerical expression for VA.
A1	Correct numbers in their choice of method.
A1	Correct length obtained.
	$AV = 16$ w/o working scores M1A1A1

Type MS · Angle between two sloping faces with the cosine rule

Answer 4 of 9 · Question 11 part e · 3 marks

2019 JUN · Paper 02 · extract 2 of 3 · find part e in this mark scheme extract

Question Number	Scheme	Marks
(c)		
M1	Use Pythagoras with a minus sign (seen or implied) or trigonometry in ΔADX OR any other complete valid method **NB**: triangle ΔADV is equilateral.	
A1	Correct numbers in their choice of method.	
A1	Correct **exact** length for the perpendicular.	
(d)		
M1	Identify the correct triangle needed with the required angle marked (may be on Figure 1). This may be shown explicitly or implied by their work that follows.	
A1	Reach one of $\tan\theta = \frac{8\sqrt{2}}{8}$, $\sin\theta = \frac{8\sqrt{2}}{8\sqrt{3}}$, $\cos\theta = \frac{8}{8\sqrt{3}}$ oe	
A1	Correct answer, **must be 1 dp.**	
(e)		
M1	Use cosine rule: $\cos\phi = \frac{"DX"^2 + "XB"^2 - "BD"^2}{2\times"DX"\times"XB"}$ (their values)	
A1	Correct numbers substituted, follow through their previous answers	
A1	Correct answer, **must be 1 dp unless already penalised in (d)**	
	Any other routes should be marked: M1 Correct, complete method (ie it must be possible to reach a value for the required angle) A1 Correct numbers substituted A1 Correct answer, **must be 1 dp unless already penalised in (d)**	

Type MS · Angle between two sloping faces with the cosine rule

Answer 4 of 9 · Question 11 part e · 3 marks

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Type MS · Angle between two sloping faces with the cosine rule

Answer 5 of 9 · Question 4 part c · 5 marks

2022 JAN · Paper 02 · extract 1 of 3 · find part c in this mark scheme extract

4(a)	$FC = \sqrt{10^2 + 10^2 + 10^2} = \sqrt{300}$	M1A1 [2]
(b)	$\cos FCA = \dfrac{\sqrt{200}}{\sqrt{300}} \Rightarrow \angle FCA = 35.3^\circ$	M1A1 [2]
(c)	$CX = \dfrac{\sqrt{200}}{2} = \sqrt{50}$, $FX = \sqrt{50+100} = \sqrt{150}$ $\cos \angle FXC = \dfrac{'150'+'50'-'300'}{2\times\sqrt{'150'}\times\sqrt{'50'}} \Rightarrow \angle FXC = 125.2643...^\circ \Rightarrow$ awrt 125°	M1,M1A1 M1A1 [5]
	Total	**9 marks**

Part	Mark	Notes
(a)	M1	For using Pythagoras theorem or any appropriate trigonometry to find FC. $FC = \sqrt{10^2 + 10^2 + 10^2} = ...$ OR $AC = \sqrt{10^2 + 10^2} = \sqrt{200}$ $FC = \sqrt{10^2 + 200} = \sqrt{300}$
	A1	For the correct exact length of FC = $\sqrt{300}\left(=10\sqrt{3}\right)$
(b)	M1	For using any appropriate trigonometry to find the required angle. They must find a value for the award of this mark. $\cos \angle FCA = \dfrac{\sqrt{200}}{\sqrt{300}} \Rightarrow \angle FCA = \left(35.2643^\circ\right)$, $\sin \angle FCA = \dfrac{10}{\sqrt{300}} \Rightarrow \angle FCA = \left(35.2643^\circ\right)$, $\tan \angle FCA = \dfrac{10}{\sqrt{200}} \Rightarrow \angle FCA = \left(35.2643^\circ\right)$ Allow awrt 14.1 for $\sqrt{200}$ and awrt 17.3 for $\sqrt{300}$
	A1	$\angle FCA = 35.2643...^\circ \approx 35.3^\circ$ (awrt) Accept answers which round to 35.3°, but they must round to this value.

Type MS · Angle between two sloping faces with the cosine rule

Answer 5 of 9 · Question 4 part c · 5 marks

2022 JAN · Paper 02 · extract 2 of 3 · find part c in this mark scheme extract

(c)	M1	Let X be the midpoint of BH. For any method (Pythagoras or trigonometry) to find the length CX $CX = \frac{\sqrt{200}}{2} = \left(\sqrt{50} \text{ or } 5\sqrt{2}\right)$
	M1	For any method (Pythagoras or trigonometry) to find the length FX $FX = \sqrt{50+100} = \left(\sqrt{150} \text{ or } 5\sqrt{6}\right)$ or $FX = \sqrt{200-50} = \left(\sqrt{150} \text{ or } 5\sqrt{6}\right)$
	A1	For both correct lengths $\sqrt{50}$ and $\sqrt{150}$
	M1	For the correct cosine rule to find angle FXC using their lengths $\cos\angle FXC = \frac{'150'+'50'-'300'}{2\times\sqrt{'150'}\times\sqrt{'50'}} \Rightarrow \angle FXC = \left(125.2643...^{\circ}\right)$ NB: Check that the cosine rule they are using is to find the required angle.
	A1	For the correct angle with awrt. $\angle FXC = 125^{\circ}$ Accept answers which round to 125°, but they must round to this value.
	ALT – Uses the right-angled triangle *FGX*	
	M1	Let X be the midpoint of BH. For any method (Pythagoras or trigonometry) to find the length GX $GX = \frac{\sqrt{200}}{2} = \left(\sqrt{50} \text{ or } 5\sqrt{2} \text{ or awrt } 7.07\right)$ If they use the sin ratio to find the angle, identify $FG = 10$ (cm)
	M1	For any method (Pythagoras or trigonometry) to find the length FX $FX = \sqrt{50+100} = \left(\sqrt{150} \text{ or } 5\sqrt{6} \text{ awrt } 12.2\right)$ or $FX = \sqrt{200-50} = \left(\sqrt{150} \text{ or } 5\sqrt{6} \text{ awrt } 12.2\right)$ **OR** If they use the tan ratio to find the angle, identify $FG = 10$ (cm)
	A1	For both correct lengths $\sqrt{50}$ and 10 for tan **or** $\sqrt{50}$ and $\sqrt{150}$ for cos **or** $\sqrt{150}$ and 10 for sin Allow decimal equivalents, i.e., 7.07 and 12.2
	M1	$\cos\angle FXG = \frac{\sqrt{50}}{\sqrt{150}} \Rightarrow \angle FXG = \left(54.7356...^{\circ}\right)$, $\sin\angle FXG = \frac{10}{\sqrt{150}} \Rightarrow \angle FXG = \left(54.7356...^{\circ}\right)$, $\tan\angle FXG = \frac{10}{\sqrt{50}} \Rightarrow \angle FXG = \left(54.7356...^{\circ}\right)$ So the required angle is $\angle FXC = 180 - '54.7356...'^{\circ} = \left(125.264...^{\circ}\right)$ We must see this calculation for the award of this mark. Do **not** award just for 54.7356…°
	A1	For the correct angle with awrt. $\angle FXC = 125^{\circ}$ Accept answers which round to 125°, but they must round to this value.

Type MS · Angle between two sloping faces with the cosine rule

Answer 5 of 9 · Question 4 part c · 5 marks

2022 JAN · Paper 02 · extract 3 of 3 · find part c in this mark scheme extract

Question	Scheme	Mark

Type MS · Angle between two sloping faces with the cosine rule

Answer 6 of 9 · Question 10 part d · 4 marks

2024 JUN · Paper 02 (regional R) · extract 1 of 3 · find part d in this mark scheme extract

10(a)	$AC = \sqrt{(2x)^2 + x^2} = (\sqrt{5}x)$ $VA = '\sqrt{5}x' \sin 45^o = \frac{\sqrt{10}}{2}x$ *	B1 M1A1 cso [3]
(b)	Height $= \sqrt{\left(\frac{\sqrt{10}}{2}x\right)^2 - \left(\frac{\sqrt{5}}{2}x\right)^2} = \frac{\sqrt{5}}{2}x$	M1A1 [2]
(c)	$\cos \angle VBA = \frac{1}{\frac{\sqrt{10}}{2}}$ $\Rightarrow \angle VBA = 50.768...^o \approx 50.8^o$ awrt	M1 A1 [2]
(d)	Let M be the point at which the diagonals AC and BD meet. The required angle is either AMB or DMC AM = MB $= \frac{AC}{2} = \frac{\sqrt{5}x}{2}$ or DM=MC=$\frac{AC}{2} = \frac{\sqrt{5}x}{2}$ $\cos \angle AMB = \frac{\left(\frac{\sqrt{5}}{2}\right)^2 + \left(\frac{\sqrt{5}}{2}\right)^2 - 2^2}{2 \times \frac{\sqrt{5}}{2} \times \frac{\sqrt{5}}{2}}$ oe $(\cos \angle AMB = -\frac{3}{5} \Rightarrow \angle AMB = 126.869...^o) \approx 126.9^o$ awrt	B1 B1 M1 A1 [4]
(e)	$9\sqrt{5} = \frac{1}{3} \times 2x \times x \times "\frac{\sqrt{5}}{2}x" \Rightarrow x = \cdots$ $x = 3$	M1 A1 [2]
		Total 13 marks

Type MS · Angle between two sloping faces with the cosine rule

Answer 6 of 9 · Question 10 part d · 4 marks

2024 JUN · Paper 02 (regional R) · extract 2 of 3 · find part d in this mark scheme extract

Part	Mark	Notes
(a)	B1	For using correct Pythagoras theorem to find length *AC or ½ AC*, explicit substitution must be seen.
	M1	For correct trigonometry using their AC in triangle VAC or half of their AC in triangle VAM or correct Pythagoras theorem (let M be the centre of the base at which the diagonals *AC* and *BD* meet) to find length *VA*. *e.g.* $VA = \sqrt{5}x\cos(45°) = \frac{\sqrt{10}}{2}x$ *or* $VA = \frac{\sqrt{5}}{2}x \div \cos(45°) = \frac{\sqrt{10}}{2}x$ *or* $VA = \frac{\sqrt{5}}{2}x \div \sin(45°) = \frac{\sqrt{10}}{2}x$
	A1 cso	Achieves printed answer with no errors. Condone AV instead of VA Note: $\frac{\sqrt{10}}{2}x$ *written as* $\frac{\sqrt{10x}}{2}$ *is A0*, $\frac{\sqrt{10}}{2}x$ *written as* $\frac{\sqrt{10}x}{2}$ *is A1*
(b)	M1	Uses the correct Pythagoras theorem or correct trigonometry with the given length VA= $\frac{\sqrt{10}}{2}x$ to find the height of the pyramid.
	A1	For $h = \frac{\sqrt{5}}{2}x$ Correct exact answer only scores M1A1. This can be written down as the triangle is isosceles.
(c)	M1	**Ignore missing *x* consistently throughout their solution.** For using any appropriate trigonometry to find the angle *VBA* If attempts cosine rule, look for correct application of the cosine rule for the required angle: $\left(\frac{\sqrt{10}}{2}x\right)^2 = \left(\frac{\sqrt{10}}{2}x\right)^2 + (2x)^2 - 2\times\frac{\sqrt{10}}{2}x\times 2x\cos(VBA)$ oe
	A1	For the correct angle. awrt 50.8 ° (with or without degree sign)
(d)	B1	Identifies the angle required.(can be implied/embedded in their working)
	B1	For the correct lengths AM and MB **or** DM and MC **or** BE and ME Let E to be mid point of AB, correct length BE=*x*, ME=0.5*x* scores B1, this mark can be implied/embedded in their working)
	M1	**Ignore missing *x* consistently throughout their solution.** For using any appropriate trigonometry to find the required angle or half of the required angle.
	A1	For the correct angle. awrt 126.9° (with or without degree sign)
(e)	M1	For using the correct formula for the volume of a pyramid with their height of the pyramid and attempts to find a value of *x*.
	A1	For $x = 3$ with or without unit, ignore incorrect unit

Type MS · Angle between two sloping faces with the cosine rule

Answer 6 of 9 · Question 10 part d · 4 marks

2024 JUN · Paper 02 (regional R) · extract 3 of 3 · find part d in this mark scheme extract

Question	Scheme	Marks

Type MS · Angle between two sloping faces with the cosine rule

Answer 7 of 9 · Question 11 part c · 6 marks

2025 NOV · Paper 02 · extract 1 of 5 · find part c in this mark scheme extract

11(a)	Let the intersection of AC and DB be X $[AC =]\sqrt{6^2+6^2}\left[=\sqrt{72} \Rightarrow AX = 3\sqrt{2}\right]$ oe or $[AX =]\sqrt{3^2+3^2}\left[=\sqrt{18}=3\sqrt{2}\right]$ $[h =]\sqrt{12^2-"\left(3\sqrt{2}\right)"^2} = 3\sqrt{14}$ oe*	M1 M1A1 cso [3]
ALT1	Let Z be the midpoint of AB $[EZ =]\sqrt{12^2-3^2}\left[=\sqrt{135}=3\sqrt{15}\right]$ oe $[h =]\sqrt{\left("3\sqrt{15}"\right)^2-(3)^2} = 3\sqrt{14}$ oe*	M1 M1A1 cso [3]
ALT2	$[h =]\sqrt{3^2+3^2+h^2} = 12$ or $3^2+3^2+h^2 = 144$	M1M1 A1
(b)	Required angle $= [\angle EAX]\sin^{-1}\left(\frac{3\sqrt{14}}{12}\right)$ or $\tan^{-1}\left(\frac{3\sqrt{14}}{3\sqrt{2}}\right)$ or $\cos^{-1}\left(\frac{3\sqrt{2}}{12}\right) = 69.295...^\circ$ or $\cos^{-1}\left(\frac{\left(6\sqrt{2}\right)^2+12^2-12^2}{2\times 6\sqrt{2}\times 12}\right) \approx 69.3^\circ$	M1A1 [2]

Type MS · Angle between two sloping faces with the cosine rule

Answer 7 of 9 · Question 11 part c · 6 marks

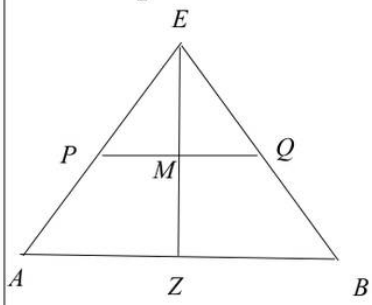
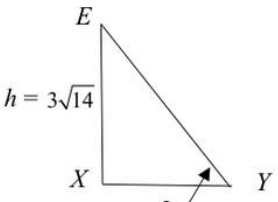
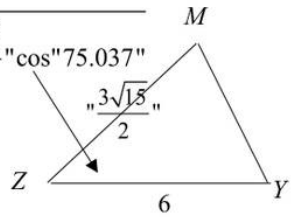
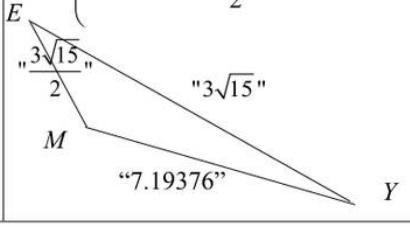
2025 NOV · Paper 02 · extract 2 of 5 · find part c in this mark scheme extract

Part	Mark	Notes
(a)	M1	For correctly applying Pythagoras theorem to find AC or AX Full method, must include square rooting.
	M1	For correctly applying Pythagoras theorem to find h Allow follow through of their AX/AC. Full method, must include square rooting. Not a dependent method mark, but must see M1 or be correctly using $3\sqrt{2}$ M0 M1 A0 will occur. We will permit eg $3\sqrt{2}^2$ if the intention is clear
	A1*cso	For the correct length only. Minimum steps shown, no errors or omissions.
ALT1	M1	For correct applying Pythagoras theorem to find EZ Full method, must include square rooting.
	M1	For correctly applying Pythagoras theorem to find h Allow follow through of their EZ. Full method, must include square rooting. Not a dependent method mark, but must see M1 or be correctly using $3\sqrt{15}$ M0 M1 A0 will occur. We will permit eg $3\sqrt{15}^2$ if the intention is clear
	A1*cso	For the correct length only. Minimum steps shown, no errors or omissions.
ALT2	M1M1	For a fully correct statement using 3D Pythagoras, each side shown. M0 M0 for any part wrong
	A1*cso	For the correct length only. Minimum steps shown, no errors or omissions.
(b)	M1	For using the given value of h and applying correct trigonometry to find the required angle. Allow ft of $3\sqrt{2}$ if from correct method. If any other method used, the **method** must be complete and correct ie leaving as sin = is not complete. Note equivalent angles (because of symmetry) eg $\angle EBD$
	A1	For the correct angle awrt 69.3°

Type MS · Angle between two sloping faces with the cosine rule

Answer 7 of 9 · Question 11 part c · 6 marks

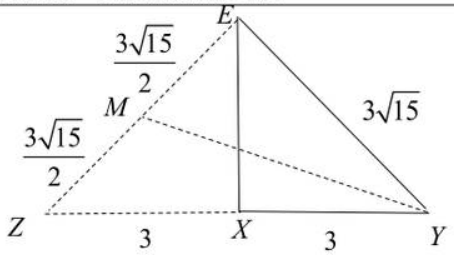
2025 NOV · Paper 02 · extract 3 of 5 · find part c in this mark scheme extract

(c)	Let M be the midpoint of PQ, Y be the midpoint of DC and Z be the midpoint of AB	
	$[EY = EZ =]\sqrt{3^2 + \left(3\sqrt{14}\right)^2}\ [= 3\sqrt{15}]$	M1
	$[EM =]\frac{"3\sqrt{15}"}{2}$	M1
	 	
	Angle between planes EAB & ABC =	
	$\left[[\angle EYX =]\tan^{-1}\left(\frac{3\sqrt{14}}{3}\right)\left[= 75.0367^\circ\right]\right.$ $\left.\text{or } \cos^{-1}\left(\frac{"\left(3\sqrt{15}\right)"^2 + 6^2 - "\left(3\sqrt{15}\right)"^2}{2\times"3\sqrt{15}"\times 6}\right) \text{ or } \cos^{-1}\left(\frac{3}{"3\sqrt{15}"}\right) \text{ or } \sin^{-1}\left(\frac{3\sqrt{14}}{"3\sqrt{15}"}\right)\right]$	M1
	$[MY =]\sqrt{6^2 + \left("\frac{3\sqrt{15}}{2}"\right)^2 - 2\times 6\times"\frac{3\sqrt{15}}{2}"\cos"75.037"}$ $= 7.19376...$ 	A1 (M1 on ePen)
	[Required angle = $\angle EMY$ =] $\cos^{-1}\left(\frac{\left("\frac{3\sqrt{15}}{2}"\right)^2 + "7.19376"^2 - \left("3\sqrt{15}"\right)^2}{2\times"\frac{3\sqrt{15}}{2}"\times"7.19376"}\right) = 126.3133 = 126$ 	dddM1 A1
		[6]

Type MS · Angle between two sloping faces with the cosine rule

Answer 7 of 9 · Question 11 part c · 6 marks

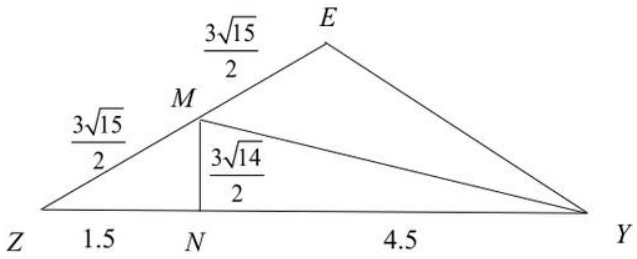
2025 NOV · Paper 02 · extract 4 of 5 · find part c in this mark scheme extract

3rd M mark and 1st A mark – ALTs to find MY		
	 $\left[\left[\angle ZEY = 2\times\angle XEY =\right] 2\times\tan^{-1}\left(\frac{3}{3\sqrt{15}}\right)\left[= 2\times 14.9632^{\circ}\right]\right]$ or $\left[\left[\angle ZEY =\right]\cos^{-1}\frac{"\left(3\sqrt{15}\right)"^2+"\left(3\sqrt{15}\right)"^2-"(6)"^2}{2\times"3\sqrt{15}"\times"\left(3\sqrt{15}\right)"}\left[= 29.9264\right]\right]$	M1
	$MY = \sqrt{\left("3\sqrt{15}"\right)^2+\left("\frac{3\sqrt{15}}{2}"\right)^2-2\times"3\sqrt{15}"\times"\frac{3\sqrt{15}}{2}"\cos 29.9264}$ $= 7.19376...$	A1
		Total 11 marks

Type MS · Angle between two sloping faces with the cosine rule

Answer 7 of 9 · Question 11 part c · 6 marks

2025 NOV · Paper 02 · extract 5 of 5 · find part c in this mark scheme extract

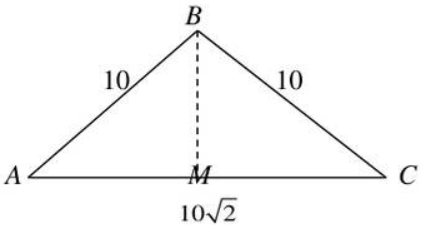
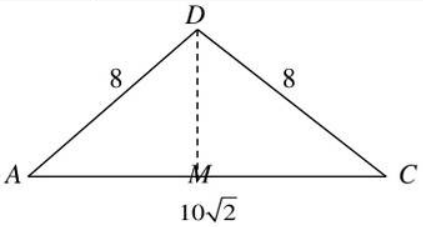
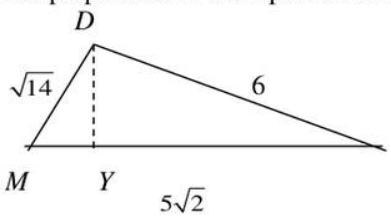
Question	Scheme	Marks
11c ALT	Dropping a perpendicular from M to the base of the pyramid, which hits at point N	
		
	$(MN=)\frac{3\sqrt{14}}{2}$ $\left(\text{Note: } MZ=\sqrt{\left(\frac{3\sqrt{14}}{2}\right)+1.5^2}=\frac{3\sqrt{15}}{2}\right)$	M1
	$(\angle ZMN=)\tan^{-1}\left(\frac{1.5}{\frac{3\sqrt{14}}{2}}\right)$ or $\cos^{-1}\left(\frac{\frac{3\sqrt{14}}{2}}{\frac{3\sqrt{15}}{2}}\right)$	M1
	or $\sin^{-1}\left(\frac{1.5}{\frac{3\sqrt{15}}{2}}\right)(=14.96321743)$	
	$(\angle YMN=)\tan^{-1}\left(\frac{4.5}{\frac{3\sqrt{14}}{2}}\right)(=38.7220711)$	M1
	$(\angle YMZ=14.96321743+38.7220711=)53.68528853$	A1
	$180-"53.68528853"=126.3$	dddM1A1
		[6]

Part	Mark	Notes
ALT	M1	For correctly stating (explicitly or in working) the height of MN
	M1	For the full and correct **method** to find $\angle ZMN$ They may follow through MZ if found and from a correct method.
	M1	For the full and correct **method** to find $\angle YMN$ They may follow through MZ if found and from a correct method.
	A1	For the correct angle $\angle YMZ$ allow awrt 39
	dddM1	Dependent on all previous method marks, for subtracting their angle from 180.
	A1	For awrt 126°

Type MS · Angle between two sloping faces with the cosine rule

Answer 8 of 9 · Question 9 part c · 4 marks

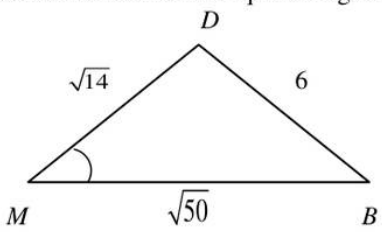
2021 NOV · Paper 02 · extract 1 of 3 · find part c in this mark scheme extract

9 (a)	$AC = \sqrt{10^2 + 10^2} = \sqrt{200} = 10\sqrt{2}$	M1A1 [2]
(b)	 $BM = \sqrt{10^2 - \left(5\sqrt{2}\right)^2} = 5\sqrt{2}$	M1A1 [2]
(c)	 $DM = \sqrt{8^2 - \left(5\sqrt{2}\right)^2} = \sqrt{14}$ $\angle DMB = \cos^{-1}\left(\frac{14 + 50 - 36}{2 \times \sqrt{14} \times 5\sqrt{2}}\right) = 58.0519...^\circ \approx 58.1^\circ$	M1A1 M1A1 [4]
(d)	Let perpendicular from point D to BM be the point Y  Vertical height of shape $ABCD$ In triangle DMY (above) Height $= DY = \sqrt{14}\sin 58.0519^\circ = 3.1749... \approx 3.17$ (cm)	M1A1 [2]
		Total 10 marks

Type MS · Angle between two sloping faces with the cosine rule

Answer 8 of 9 · Question 9 part c · 4 marks

2021 NOV · Paper 02 · extract 2 of 3 · find part c in this mark scheme extract

Part	Mark	Notes
(a)	M1	For using Pythagoras theorem correctly on triangle ABC to find AC
	A1	For $AC = 10\sqrt{2}$
(b)	M1	For using Pythagoras theorem or any appropriate trigonometry correctly on triangle ABC to find BM
	A1	For the correct length $BM = 5\sqrt{2}$ **NB:** Allow for the answer just seen without any working as the triangle ABC is isosceles.
(c)	M1	For using Pythagoras theorem to find the length DM
	A1	For $DM = \sqrt{14}$ or accept awrt 3.74
	M1	For using cosine rule to find the required angle DMB 
	A1	For awrt 58.1°
(d)	M1	For using any appropriate trigonometry to find the required length. Ft their angle DMB
	A1	For awrt 3.17 (cm) Accept 3.18 (cm)

Type MS · Angle between two sloping faces with the cosine rule

Answer 8 of 9 · Question 9 part c · 4 marks

2021 NOV · Paper 02 · extract 3 of 3 · find part c in this mark scheme extract

Question number	Scheme	Marks

Type MS · Angle between two sloping faces with the cosine rule

Answer 9 of 9 · Question 9 part d · 4 marks

2023 JAN · Paper 01 (regional R) · extract 1 of 3 · find part d in this mark scheme extract

9(a)	$AC = \sqrt{12^2+12^2} = \sqrt{288} = 12\sqrt{2}$	M1A1 [2]
(b)	$x = \dfrac{\frac{12\sqrt{2}}{2}}{\cos 30^o} = \sqrt{96} = \left(4\sqrt{6}\right)*$	M1A1 cso [2]
(c)	$\cos \angle AOB = \dfrac{96+96-12^2}{2\times\sqrt{96}\times\sqrt{96}} \Rightarrow \angle AOB = 75.522...^o$ Area $\Delta OAB = \dfrac{1}{2}\times\sqrt{96}\times\sqrt{96}\times\sin'75.522^o{}' = (46.4758...)$ $4\times 46.4758+12^2 = 329.90.... \Rightarrow$ Area = 330 [m^2]	M1A1 M1 M1A1 [5]
(d)	$\left[\angle OBC = \dfrac{180-75.522}{2} = 52.239^o\right]$ Let Y on OB be the foot of the perpendicular from A to OB $AY = 12\sin 52.239^o = (9.4868...)$ $\cos \angle AYC = \dfrac{'9.4868'^2+'9.4868'^2-288}{2\times'9.4868'\times'9.4868'} \Rightarrow \angle AYC = 126.87... \approx 127^{[o]}$	M1 A1 M1A1 [4]
		Total 13 marks

Question	Notes	Marks
9(a)	For using Pythagoras theorem on triangle ABC or triangle ADC $AC = \sqrt{12^2+12^2} = ...$	M1
	For the correct value of AC $AC = \sqrt{288} = 12\sqrt{2}$	A1 [2]
(b)	Let the intersection of AC and BD be X Uses any appropriate trigonometry on triangle OAX For example; $x = \dfrac{\frac{12\sqrt{2}}{2}}{\cos 30^o} = ...$ **OR** $\dfrac{\sin 120^o}{12\sqrt{2}} = \dfrac{\sin 30^o}{OA} \Rightarrow OA = ...$ OR $\left(12\sqrt{2}\right)^2 = x^2+x^2-2\times x\times x\cos 120 \Rightarrow x = ...$	M1
	For the correct value of x	A1 cso

Type MS · Angle between two sloping faces with the cosine rule

Answer 9 of 9 · Question 9 part d · 4 marks

2023 JAN · Paper 01 (regional R) · extract 2 of 3 · find part d in this mark scheme extract

	$x = \sqrt{96} = \left(4\sqrt{6}\right)$ *	[2]
(c)	For using trigonometry to find angle $\angle AOB$ $\cos \angle AOB = \frac{96+96-12^2}{2\times\sqrt{96}\times\sqrt{96}} = ...$	M1
	$\angle AOB = 75.522...^{\circ}$	A1
	Area of triangle OAB Area $= \frac{1}{2}\times\sqrt{96}\times\sqrt{96}\times\sin'75.522^{\circ}' = \left(46.4758...\right)$	M1
	Total area of the pyramid $= 4\times46.4758+12^2 = \left(329.90....\right)$	M1
	For the correct final area = awrt 330 (cm^2)	A1 [5]
	ALT	
	Height of the triangle of one of the triangular faces: $h = \sqrt{\left(4\sqrt{6}\right)^2 - 6^2} = ...$	M1
	$h = 2\sqrt{15}$	A1
	Area of triangle $OAB = \frac{1}{2}\times2\sqrt{15}\times12 = 12\sqrt{15} = \left(46.4758...\right)$	M1
	Total area of the pyramid $= 4\times12\sqrt{15}+144 = \left(329.903...\right)$	M1
	For the correct final area = awrt 330 (cm^2) Allow an exact answer of $48\sqrt{15}+144$ $(cm)^2$ oe	A1
(d)	$\cos \angle AOB = \frac{96+96-12^2}{2\times\sqrt{96}\times\sqrt{96}} = 75.522^{\circ}$ or $\angle OBC = \frac{180^{\circ}-75.522^{\circ}}{2} = 52.239^{\circ}$	
	Let Y on OB be the foot of the perpendicular from A to OB Length $AY = 12\sin52.239^{\circ} = \left(9.4868...\right)$ OR Length $AY = 4\sqrt{6}\sin75.522^{\circ} = \left(9.4868...\right)$	M1A1 [M1A1]
	For the appropriate trigonometry on triangle AYC to find angle AYC $\cos \angle AYC = \frac{'9.4868'^2+'9.4868'^2-288}{2\times'9.4868'\times'9.4868'} \Rightarrow \left(\angle AYC = 126.87\right)$	M1
	Angle between plane AOB and plane OBC =awrt 127[°]	A1 [4]
		Total 13 marks

Type MS · Angle between two sloping faces with the cosine rule

Answer 9 of 9 · Question 9 part d · 4 marks

2023 JAN · Paper 01 (regional R) · extract 3 of 3 · find part d in this mark scheme extract

Question	Scheme	Marks